

WATER POLLUTION

PYQs

Last 3 Years Detailed PYQ Solution (2022-2024)

2 SEPTEMBER 2024

Chlorination is most common disinfection process in swimming pools. Which of the chemical causes respiratory and eye irritation around swimming pools immediately after disinfection process?

1. Hypochlorous acid
2. Chlorine gas
3. Chloramines
4. Chlorine dioxide gas

ANSWER: C

1. Chlorine in Pools:

- When chlorine is added to pool water, it typically forms hypochlorous acid (HOCl) and hypochlorite ions (OCl-). Both of these compounds are effective disinfectants, killing bacteria and other pathogens to keep the pool safe for swimmers.

2. Formation of Chloramines:

- Chloramines are formed when chlorine reacts with nitrogen-containing compounds, which are commonly introduced into the pool from swimmers' sweat, urine, skin cells, and other organic matter.
- The reaction between chlorine and these compounds produces different types of chloramines, including monochloramine (NH2Cl), dichloramine (NHCl2), and trichloramine (NCl3).

3. Chloramines and Irritation:

- **Monochloramine** and **dichloramine** can cause a mild level of irritation, but **trichloramine** is particularly notorious for causing significant respiratory and eye irritation.
- Trichloramine is volatile, meaning it can easily become airborne and be inhaled, leading to irritation of the mucous membranes in the eyes, nose, and throat.

Which of the following statements are correct?

A. BOD curve in lake water follows 1st order kinetic

B. Ultimate BOD increases in direct proportion to concentration of degradable organic matter

C. Oxygen depletion in lake water is related to both ultimate BOD and the rate constant (R)

D. Numerical value of BOD rate constant does not depend on water temperature

E. Numerical value of BOD rate constant does not depend on nature of the waste

Choose the most appropriate answer from the options given below:

1. A, B and D Only

2. B, C, D and E Only

3. A, B and C Only

4. A, B, C and D Only

ANSWER: C

Let's analyze each statement to determine which are correct:

A. BOD curve in lake water follows 1st order kinetic:

- This statement is correct. The Biochemical Oxygen Demand (BOD) curve typically follows first-order kinetics, meaning the rate of oxygen consumption is proportional to the amount of degradable organic matter remaining.

B. Ultimate BOD increases in direct proportion to concentration of degradable organic matter:

- This statement is also correct. The ultimate BOD (L) is directly proportional to the concentration of degradable organic matter present in the water.

C. Oxygen depletion in lake water is related to both ultimate BOD and the rate constant (R):

- This statement is correct. Oxygen depletion is influenced by both the ultimate BOD and the rate constant (k), which together determine the rate at which oxygen is consumed.

D. Numerical value of BOD rate constant does not depend on water temperature:

- This statement is incorrect. The BOD rate constant (k) is temperature-dependent and typically increases with increasing temperature.

E. Numerical value of BOD rate constant does not depend on nature of the waste:

- This statement is also incorrect. The BOD rate constant (k) does depend on the nature of the waste, as different organic materials have varying rates of biodegradation.

So, the correct answers are: **A, B, and C**.

Which of the following statements are correct?

A. Arsenic is used in wood preservative.

B. CrO_4^2 is water soluble.

C. Arsenic is not a metal.

D. Fe III is more soluble compared to Fe II in pure water.

E. PbS and $PbCO_3$ are highly insoluble in water.

Choose the most appropriate answer from the options given below:

1. A and C Only
2. B, C, D and E Only
3. C and D Only
4. A, B, C and E Only

ANSWER: D

Let's evaluate each statement:

A. Arsenic is used in wood preservative:

- Correct. Arsenic has been used in wood preservatives, notably in compounds such as chromated copper arsenate (CCA).

B. CrO_4^{2-} (chromate ion) is water soluble:

- Correct. Chromate salts are generally water-soluble.

C. Arsenic is not a metal:

- Correct. Arsenic is a metalloid, which means it has properties of both metals and non-metals.

D. Fe^{3+} (Ferric iron) is more soluble compared to Fe^{2+} (Ferrous iron) in pure water:

- Incorrect. Fe^{2+} (Ferrous iron) is generally more soluble in water compared to Fe^{3+} (Ferric iron).

E. PbS (Lead sulfide) and $PbCO_3$ (Lead carbonate) are highly insoluble in water:

- Correct. Both PbS and $PbCO_3$ are highly insoluble in water.

Based on this analysis, the correct answer is: **4. A, B, C and E Only**.

Fe^{2+} (Ferrous iron) is generally more soluble in water compared to Fe^{3+} (Ferric iron) due to several factors related to their chemical properties:

1. **Charge and Ionic Size**:
    - Fe^{2+} has a +2 charge, whereas Fe^{3+} has a +3 charge. The higher charge density on Fe^{3+} leads to a stronger attraction to other negatively charged species, making it more likely to form insoluble compounds.
2. **Hydrolysis and Precipitation**:
    - Fe^{3+} tends to undergo hydrolysis in water, forming $Fe(OH)_3$, which is highly insoluble and precipitates out of the solution. Fe^{2+}, on the other hand, forms $Fe(OH)_2$, which is relatively more soluble in water.
3. **Redox Chemistry**:
    - Fe^{2+} is more stable under reducing conditions, while Fe^{3+} is more stable under oxidizing conditions. In natural water systems, the presence of reducing agents can keep Fe^{2+} in solution, while Fe^{3+} is more prone to precipitation.
4. **Complexation**:
    - Fe^{2+} can form soluble complexes with various ligands in water, such as chloride or sulfate ions, which enhances its solubility. Fe^{3+} also forms complexes, but these are often less soluble due to the higher charge and tendency to hydrolyze.

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Identify the correct sequence of the following units in the sewage treatment scheme of a liquid stream:

(A) Trickling filter

(B) Primary settling tank

(C) Grit chamber

(D) Screen chamber

(E) Secondary settling tank

Choose the correct answer from the options given below:

(1) (D), (A), (B), (C), (E)

(2) (D), (C), (B), (A), (E)

(3) (C), (A), (D), (B). (E)

(4) (A), (E), (D), (C), (B)

ANSWER: B

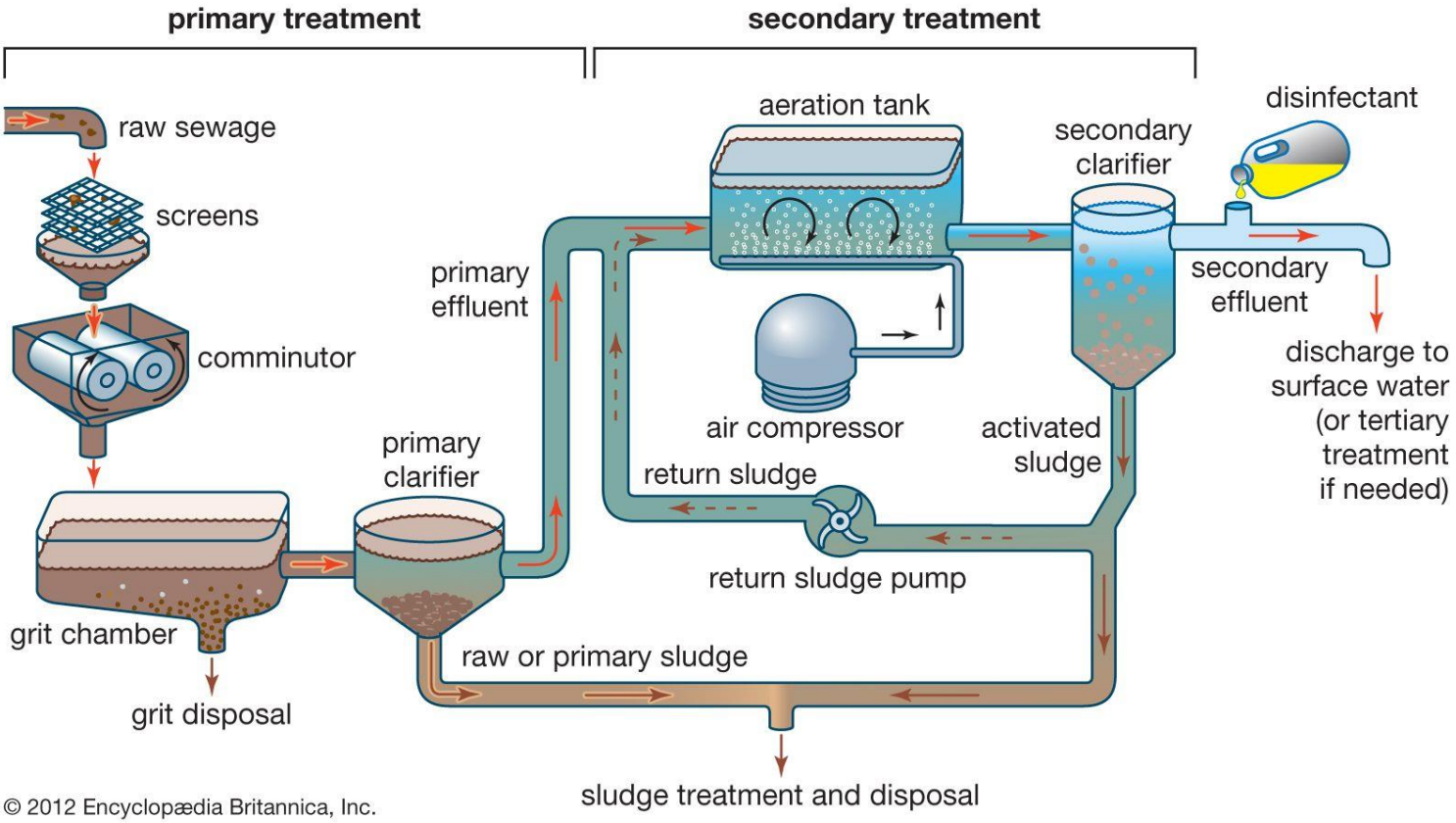
To identify the correct sequence of units in the sewage treatment scheme for the liquid stream, let us review the typical steps involved:

1. **Screen chamber**: Removes large debris and solid particles from sewage.
2. **Grit chamber**: Removes heavier inorganic particles (like sand and gravel).
3. **Primary settling tank**: Allows heavier organic solids to settle out of the liquid.
4. **Trickling filter**: Provides biological treatment by allowing microorganisms to break down organic matter.

5. **Secondary settling tank**: Removes suspended biological solids after the trickling filter.

Based on this typical sequence, the correct answer is:

(2) (D), (C), (B), (A), (E)



Arrange the following membrane processes in order of decreasing pore size :

(A) Nanofiltration

(B) Microfiltration

(C) Ultrafiltration

(D) Hyperfiltration

Choose the correct answer from the options given below:

(1) (B), (C), (A), (D)

(2) (A), (B), (C), (D)

(3) (D), (A), (B), (C)

(4) (C), (A), (B), (D)

ANSWER: A

Membrane Process	Pore Size Range	Order
Microfiltration (B)	0.1 to 10 micrometers	Largest
Ultrafiltration (C)	0.01 to 0.1 micrometers	Second
Nanofiltration (A)	0.001 to 0.01 micrometers	Third
Hyperfiltration (D)	<0.001 micrometers	Smallest

Which of the following DO NOT contribute to hardness of water ?

(A) K^+

(B) Na^+

(C) Ca^{2+}

(D) Mg^{2+}

(E) Fe^{3+}

Choose the correct answer from the options given below :

(1) (A) and (B) Only

(2) (C) and (D) Only

(3) (B), (C) and (D) Only

(4) (A), (B) and (E) Only

ANSWER: A

Ion	Contribution to Hardness	Reason
K^+	None	Monovalent cation, does not form insoluble salts
Na^+	None	Monovalent cation, does not form insoluble salts
Ca^{2+}	High	Divalent cation, forms insoluble compounds (e.g., $CaCO_3$)
Mg^{2+}	High	Divalent cation, forms insoluble compounds (e.g., $MgCO_3$)
Fe^{3+}	Moderate	Forms insoluble compounds (e.g., $Fe(OH)_3$)

Match List - I with List - II.

LIST-I Water Contaminants		**LIST-II** Preferred Removal Method	
A.	Mn2+	**I.**	Activated Carbon
B.	Ca2+ and HCO3	**II.**	Raise pH by adding Na2CO3
C.	Trihalomethanes	**III.**	Addition of lime
D.	Mg2+	**IV.**	Oxidation

Choose the correct answer from the options given below :

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(3) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(4) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

ANSWER: A

Let's match the water contaminants with their preferred removal methods:

A. Mn^{2+} (Manganese) - IV. Oxidation: Manganese is typically removed from water by oxidation, which converts soluble Mn^{2+} to insoluble MnO_2, which can then be filtered out.

B. Ca^{2+} and HCO_3^- (Calcium and Bicarbonate) - III. Addition of lime: This process is commonly used in water softening, where the addition of lime ($Ca(OH)_2$) helps to precipitate calcium carbonate ($CaCO_3$) and remove hardness.

C. Trihalomethanes - I. Activated Carbon: Trihalomethanes (THMs) are removed using activated carbon adsorption, which effectively traps these compounds on its surface.

D. Mg^{2+} (Magnesium) - II. Raise pH by adding Na_2CO_3: Magnesium hardness can be reduced by raising the pH with sodium carbonate (Na_2CO_3), leading to the precipitation of magnesium carbonate ($MgCO_3$).

Based on this matching, the correct answer is:

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

The volume of sludge in the digester divided by the volume of digested sludge withdrawn each day in an anaerobic process is known as:

(1) hydraulic retention time

(2) solid retention time

(3) total volatiles

(4) slurry density

ANSWER: A

The term you are looking for is **hydraulic retention time (HRT)**.

Hydraulic retention time refers to the length of time that liquid (or sludge) remains in the digester, calculated as the volume of sludge in the digester divided by the volume of digested sludge withdrawn each day. It is a critical parameter in the design and operation of anaerobic digesters as it influences the efficiency of the digestion process.

Let's dive into the details of the hydraulic retention time (HRT) in an anaerobic digestion process:

Hydraulic Retention Time (HRT)

Definition: Hydraulic retention time (HRT) is the average time that liquid (or sludge) spends in a digester. It's a crucial parameter in anaerobic digestion because it affects the extent of the treatment process and the efficiency of biogas production.

Calculation:

$$\text{HRT} = \frac{\text{Volume of sludge in the digester}}{\text{Volume of digested sludge withdrawn each day}}$$

Anaerobic Digestion Process

Anaerobic digestion is a series of biological processes in which microorganisms break down organic matter in the absence of oxygen, producing biogas (mainly methane and carbon dioxide) and digestate (a nutrient-rich substance). The process occurs in several stages:

1. **Hydrolysis**:
    - Complex organic matter (carbohydrates, proteins, and fats) is broken down into simpler soluble compounds (sugars, amino acids, and fatty acids) by hydrolytic bacteria.
2. **Acidogenesis**:
    - Hydrolyzed compounds are converted into volatile fatty acids, hydrogen, and carbon dioxide by acidogenic bacteria.
3. **Acetogenesis**:
    - Volatile fatty acids are further broken down into acetic acid, hydrogen, and carbon dioxide by acetogenic bacteria.
4. **Methanogenesis**:
    - Methanogenic bacteria convert acetic acid, hydrogen, and carbon dioxide into methane and water. This stage is crucial for biogas production.

Role of HRT in Anaerobic Digestion

HRT is vital for ensuring that the microbial populations have sufficient time to break down the organic matter and produce biogas. The HRT affects several factors:

1. **Efficiency**:
    - Longer HRT allows for more complete degradation of organic matter, resulting in higher biogas yield and better-quality digestate.
2. **Microbial Activity**:
    - Different microbial groups thrive at different stages of the digestion process. Adequate HRT ensures a balanced microbial ecosystem within the digester.

3. **Process Stability**:
    - Maintaining an appropriate HRT helps prevent process instability and issues such as acidification, which can inhibit methanogenesis and reduce biogas production.
4. **Volume of Digester**:
    - HRT influences the required volume of the digester. Shorter HRTs require larger volumes to achieve the same level of treatment as longer HRTs.

Which of the following is NOT a design parameter in flocculent settling ?

(1) Surface over flow rate

(2) Depth of tank

(3) Hydraulic retention time

(4) Chemical coagulant dose

ANSWER: D

In the context of flocculent settling, the design parameters typically include surface overflow rate, depth of tank, and hydraulic retention time. These parameters influence the settling behavior and the design of the sedimentation tank.

However, the **chemical coagulant dose** is not a design parameter for the settling process itself. Instead, it is a parameter related to the chemical treatment process that precedes or accompanies settling.

Therefore, the correct answer is:

(4) Chemical coagulant dose

Parameter	Definition	Importance	Units/Calculation
Surface Overflow Rate	Flow rate per unit surface area of the tank	Determines hydraulic loading, affects settling efficiency	$m^3/day/m^2$ or gpd/ft^2
Depth of Tank	Vertical distance from water surface to tank bottom	Impacts detention time and settling path length	Typically in meters or feet
Hydraulic Retention Time	Average time water remains in the tank	Crucial for allowing particles to settle	

Identify the correct decreasing order of magnitude for Theoretical Oxygen Demand (ThOD), ultimate and 5-day Biochemical Oxygen Demand (BOD and BOD5) and Chemical Oxygen Demand (COD) :

(A) BOD_5

(B) BOD

(C) COD

(D) ThOD

Choose the correct answer from the options given below :

(1) (A), (B), (C), (D)

(2) (D), (B), (C), (A)

(3) (D), (B), (A), (C)

(4) (D), (C), (B), (A)

ANSWER: D

To determine the correct order, let's understand each parameter:

1. Theoretical Oxygen Demand (ThOD):

- Represents the maximum amount of oxygen required to completely oxidize the organic and inorganic compounds in a sample to carbon dioxide and water.
- Typically the highest value since it assumes complete oxidation.

2. Chemical Oxygen Demand (COD):

- Measures the total quantity of oxygen required to oxidize all organic material into carbon dioxide and water using a strong chemical oxidant.

- Usually slightly less than ThOD because not all compounds may be oxidized completely in a chemical process.

3. Ultimate Biochemical Oxygen Demand (BOD):

- Represents the total amount of oxygen consumed by microorganisms during the complete decomposition of organic matter in a sample.
- Usually lower than COD since it only measures the biodegradable portion of organic matter.

4. 5-day Biochemical Oxygen Demand (BOD5):

- Measures the amount of oxygen consumed by microorganisms during the first five days of decomposition.
- Always lower than ultimate BOD because it only accounts for the initial period of biodegradation.

Given this understanding, the correct decreasing order of magnitude is:

(4) (D), (C), (B), (A)

This means: ThOD > COD > BOD > BOD5.

The current recommended limit for perfluorooctanoic acid (PFOA), a commonly found PFAS (per-and polyfluoroalkyl substance), in drinking water as set by the US Environmental Protection Agency (EPA) is:

(1) 4 parts per trillion (ppt)

(2) 20 parts per trillion (ppt)

(3) 70 parts per trillion (ppt)

(4) 100 parts per trillion (ppt)

ANSWER: A

The current recommended limit for perfluorooctanoic acid (PFOA) in drinking water, as set by the US Environmental Protection Agency (EPA), is **4 parts per trillion (ppt)**.

So, the correct answer is:

(1) 4 parts per trillion (ppt)

Identify the coagulant aids :

(A) Clay

(B) Activated Silica

(C) Polymers

(D) Aluminium Sulfate

(E) Ferric Sulfate

Choose the correct answer from the options given below :

(1) (A) and (B) Only

(2) (C), (D) and (E) Only

(3) (A), (B) and (C) Only

(4) (D) and (E) Only

ANSWER: C

Let's identify the coagulant aids from the given options:

Coagulant Aids:

- **Clay (A)**: Often used as a coagulant aid to enhance the flocculation process.
- **Activated Silica (B)**: A common coagulant aid that improves the formation of flocs.
- **Polymers (C)**: Frequently used as coagulant aids to increase floc size and improve settling.

Primary Coagulants:

- **Aluminium Sulfate (D)**: A primary coagulant used to destabilize colloidal particles.
- **Ferric Sulfate (E)**: Another primary coagulant used for similar purposes as aluminium sulfate.

Given this information, the correct answer, indicating the substances that are coagulant aids, is:

(3) (A), (B) and (C) Only

Aspect	Primary Coagulants	Coagulant Aids
Function	Destabilize colloidal particles	Enhance coagulation and flocculation
Common Types	Aluminium Sulfate, Ferric Sulfate, Ferric Chloride	Activated Silica, Polymers, Clay
Mechanism	Neutralize particle charges to form microflocs	Improve floc formation, increase floc size
Role in Treatment	Essential for initial coagulation	Support and enhance primary coagulation

Which of the following is true for reductive dechlorination ?

(A) It is also called dehalorespiration.

(B) It is an aerobic process.

(C) Hydrogen can be used as an alternate electron acceptor.

(D) Halogenated organic compounds function as electron acceptor.

(E) It is an environmentally insignificant process.

Choose the correct answer from the options given below :

(1) (B) and (C) Only

(2) (A) and (D) Only

(3) (A) and (B) Only

(4) (C) and (E) Only

ANSWER: B

Let us analyze each statement to determine which ones are true for reductive dechlorination:

A. It is also called dehalorespiration:

- True. Reductive dechlorination is also known as dehalorespiration. It is a process where certain bacteria use halogenated organic compounds as electron acceptors.

B. It is an aerobic process:

- False. Reductive dechlorination is an anaerobic process, meaning it occurs in the absence of oxygen.

C. Hydrogen can be used as an alternate electron acceptor:

- False. Hydrogen is actually used as an electron donor in reductive dechlorination, not as an electron acceptor.

D. Halogenated organic compounds function as electron acceptor:

- True. In reductive dechlorination, halogenated organic compounds (e.g., chlorinated solvents) function as electron acceptors.

E. It is an environmentally insignificant process:

- False. Reductive dechlorination is environmentally significant as it is used for the bioremediation of contaminated sites.

Given this analysis, the correct answer is:

(2) (A) and (D) Only

Reductive Dechlorination: An Overview

Reductive dechlorination is a critical biochemical process used in the biodegradation of halogenated organic compounds, particularly chlorinated solvents. It's an anaerobic process, meaning it occurs in environments devoid of oxygen. Let's break down the process in detail:

Key Concepts

1. **Dehalorespiration**:
    - Also known as reductive dechlorination, this process involves certain anaerobic bacteria using halogenated organic compounds as electron acceptors during respiration.
    - These bacteria gain energy by reducing these compounds, typically by removing chlorine atoms and replacing them with hydrogen atoms.
2. **Electron Donors and Acceptors**:
    - **Electron Donors**: Compounds like hydrogen, organic acids, and alcohols are used as electron donors in the process.
    - **Electron Acceptors**: Halogenated organic compounds, such as tetrachloroethene (PCE) and trichloroethene (TCE), function as electron acceptors.

Mechanism of Reductive Dechlorination

1. **Initial Step**:
    - The process starts with the reduction of highly chlorinated compounds, such as PCE, into less chlorinated intermediates through the sequential removal of chlorine atoms.
    - Each dechlorination step involves the transfer of electrons from the electron donor to the halogenated compound.
2. **Intermediate Compounds**:
    - The sequential reduction of PCE yields intermediates such as trichloroethene (TCE), dichloroethene (DCE), and vinyl chloride (VC).
3. **Complete Dechlorination**:
    - The final product of complete reductive dechlorination is typically ethene, a non-toxic compound, or other harmless end-products.
    - Here's a simplified reaction sequence:
        - **PCE (C_2Cl_4) → TCE (C_2HCl_3) → DCE ($C_2H_2Cl_2$) → VC (C_2H_3Cl) → Ethene (C_2H_4)**

Environmental Significance

1. **Bioremediation**:
    - Reductive dechlorination is an essential process for the bioremediation of contaminated environments. It is used to clean up sites contaminated with chlorinated solvents, which are common industrial pollutants.
2. **Sustainable Approach**:
    - This process offers a sustainable and cost-effective method for the degradation of hazardous compounds without generating harmful by-products.
3. **Microbial Community**:

- The process relies on specific microbial communities, often including members of the genera *Dehalococcoides*, *Desulfitobacterium*, and *Dehalobacter*.
- These microorganisms possess specialized enzymes called reductive dehalogenases, which facilitate the dechlorination reaction.

Challenges and Considerations

1. **Environmental Conditions**:
   - The efficiency of reductive dechlorination is influenced by environmental factors such as pH, temperature, and the presence of other electron acceptors (e.g., nitrate, sulfate) that may compete with the halogenated compounds.
2. **Intermediate Toxicity**:
   - Some intermediates, like vinyl chloride, are more toxic than the parent compounds, necessitating careful management and monitoring of the biodegradation process.

Summary Table

Stage	**Process**	**Resulting Compound**
Initial Step	Reduction of PCE	TCE
Intermediate Step	Further reduction	DCE
Complete Dechlorination	Sequential removal of chlorine atoms	VC
Final Product	Complete reduction	Ethene

In summary, reductive dechlorination is a critical anaerobic process for the biodegradation of chlorinated organic compounds, playing a vital role in environmental remediation and sustainable waste management.

JUNE 2024 Cancelled

Choose the incorrect statement

(1) Water bodies with extensive algal bloom have as high pH as 10

(2) Removal of CO_2 by photosynthesis by algae is responsible for high pH

(3) In dark hours, algae in water bodies increases pH of water

(4) Water bodies with algal bloom experiences diurnal variation in pH

ANSWER: C

Let's analyze each statement:

(1) Water bodies with extensive algal bloom have as high pH as 10:

- Correct. Algal blooms can cause the pH of water bodies to rise significantly, sometimes reaching as high as 10 due to the removal of CO_2 through photosynthesis.

(2) Removal of CO_2 by photosynthesis by algae is responsible for high pH:

- Correct. During photosynthesis, algae remove CO_2 from the water, which reduces the concentration of carbonic acid and raises the pH.

(3) In dark hours, algae in water bodies increases pH of water:

- Incorrect. During dark hours, algae respire instead of photosynthesizing, which means they consume oxygen and produce CO_2, leading to a decrease in pH.

(4) Water bodies with algal bloom experiences diurnal variation in pH:

- Correct. Water bodies with algal blooms typically experience diurnal variations in pH, with higher pH levels during the day when photosynthesis occurs and lower pH levels at night when respiration occurs.

Given this analysis, the incorrect statement is:

(3) In dark hours, algae in water bodies increases pH of water

Which of the following is not applied/used for disinfection of water?

(1) Hypochlorous acid

(2) Chlorine dioxide

(3) Ozone

(4) Perchlorate

ANSWER: D

Perchlorate is not used for disinfection of water.

Hypochlorous acid, **chlorine dioxide**, and **ozone** are commonly used disinfectants for treating water. They effectively kill or inactivate harmful microorganisms, ensuring that water is safe for consumption.

Therefore, the correct answer is:

(4) Perchlorate

Application	Perchlorate Compound	Reason for Use
Rocket Propellants	Ammonium perchlorate (NH_4ClO_4)	Provides oxygen for fuel combustion
Explosives	Various perchlorate salts	Strong oxidizing properties
Fireworks	Potassium perchlorate ($KClO_4$)	Enables colorful combustion reactions
Chemical Synthesis	Various perchlorate compounds	Acts as reagents and catalysts
Electrochemical Applications	Various perchlorate compounds	Stability and strong oxidizing properties

Which of the phosphorus species will dominantly exist in the pH range of 3-7 in natural waters ?

(1) Insoluble Fe/Al phosphate

(2) H_2PO_4-

(3) HPO_4

(4) Insoluble Ca phosphate

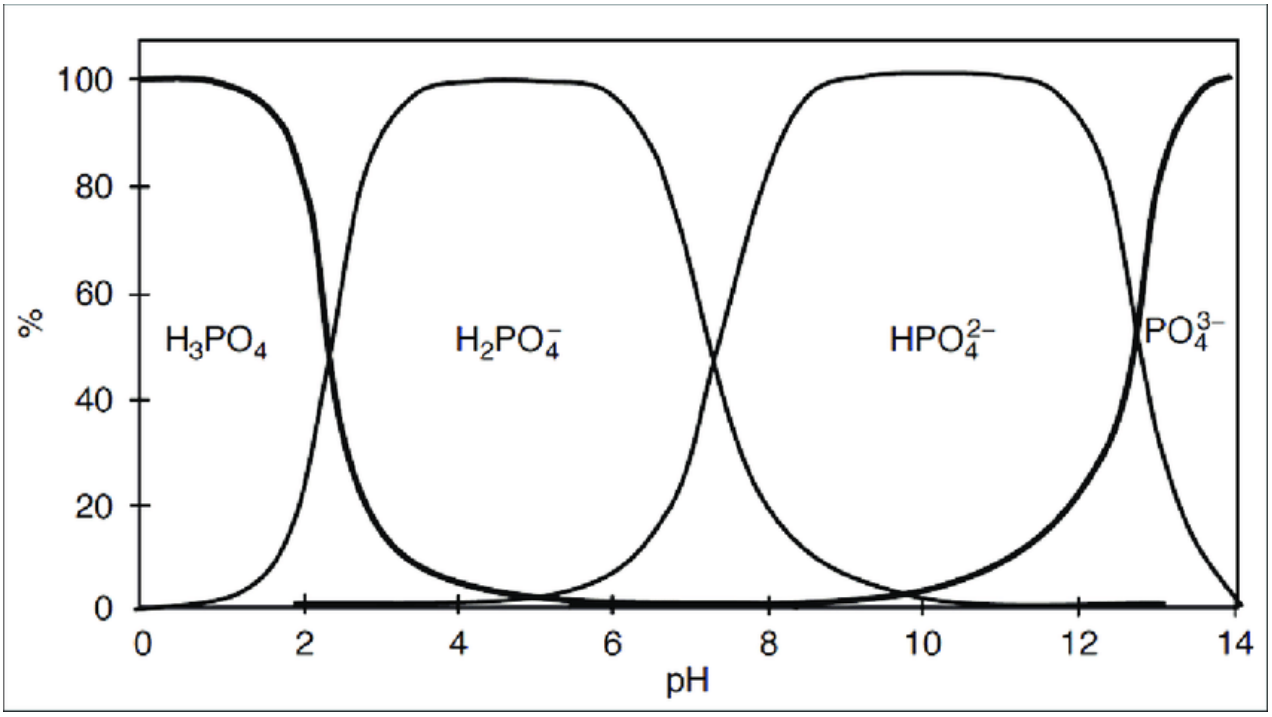
ANSWER: B

In natural waters, the dominant phosphorus species present in the pH range of 3-7 is **$H_2PO_4^-$ (dihydrogen phosphate ion)**.

Here's why:

- **$H_2PO_4^-$ (Dihydrogen phosphate ion)**: In the pH range of 3-7, $H_2PO_4^-$ is the predominant species. At lower pH values, it can coexist with H_3PO_4 (phosphoric acid), but within this range, $H_2PO_4^-$ is more dominant.
- **HPO_4^{2-} (Hydrogen phosphate ion)**: This species becomes more prevalent at higher pH levels, generally above 7.
- **Insoluble Fe/Al phosphate** and **Insoluble Ca phosphate**: These species are not dominant in dissolved form within the pH range of 3-7. They tend to precipitate out of solution at certain pH levels but do not represent the primary dissolved phosphorus species in this pH range.

So, the correct answer is: **(2) $H_2PO_4^-$**


100
80
60
40
20
0
%
H_3PO_4
$H_2PO_4^-$
HPO_4^{2-}
PO_4^{3-}
0
2
4
6
8
10
12
14
pH

Choose the incorrect statement:

(1) O_2 dissolved in surface water does not react chemically with water molecules

(2) Carbon dioxide is insoluble in boiling water

(3) Removal of CO_2 increases pH of water and changes alkalinity from carbonate to hydroxide

(4) Addition of CO_2 in water increases CaCO, formation and precipitation

ANSWER: B

Let's evaluate each statement:

(1) O_2 dissolved in surface water does not react chemically with water molecules:

- Correct. Dissolved oxygen (O_2) in water does not chemically react with water molecules (H_2O).

(2) Carbon dioxide is insoluble in boiling water:

- Incorrect. Carbon dioxide is less soluble in boiling water compared to cold water, but it is not completely insoluble. Some amount of CO_2 can still dissolve in boiling water.

(3) Removal of CO_2 increases pH of water and changes alkalinity from carbonate to hydroxide:

- Correct. Removing CO_2 from water reduces carbonic acid, thereby increasing the pH and shifting the alkalinity balance towards hydroxide ions.

(4) Addition of CO_2 in water increases $CaCO_3$ formation and precipitation:

- Correct. Addition of CO_2 can lead to the formation of carbonic acid, which in turn reacts with calcium ions to form calcium carbonate ($CaCO_3$) and causes its precipitation.

Given this analysis, the incorrect statement is:

(2) Carbon dioxide is insoluble in boiling water

December 2023

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as

Assertion (A): Chlorine dioxide has certain advantages over free chorine and hypochlorite in water disinfection.

Reason (R): Chlorine dioxide is an unstable gas and does not combine with ammonia or other natural organics to produce chloramines or chloroform.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

ANSWER: A

Let's analyze the Assertion (A) and Reason (R) statements:

Assertion (A): Chlorine dioxide has certain advantages over free chlorine and hypochlorite in water disinfection.

- **Correct**. Chlorine dioxide is known to be effective in water disinfection with several advantages, such as a broad spectrum of microbial inactivation, fewer harmful by-products, and better performance in the presence of organic matter.

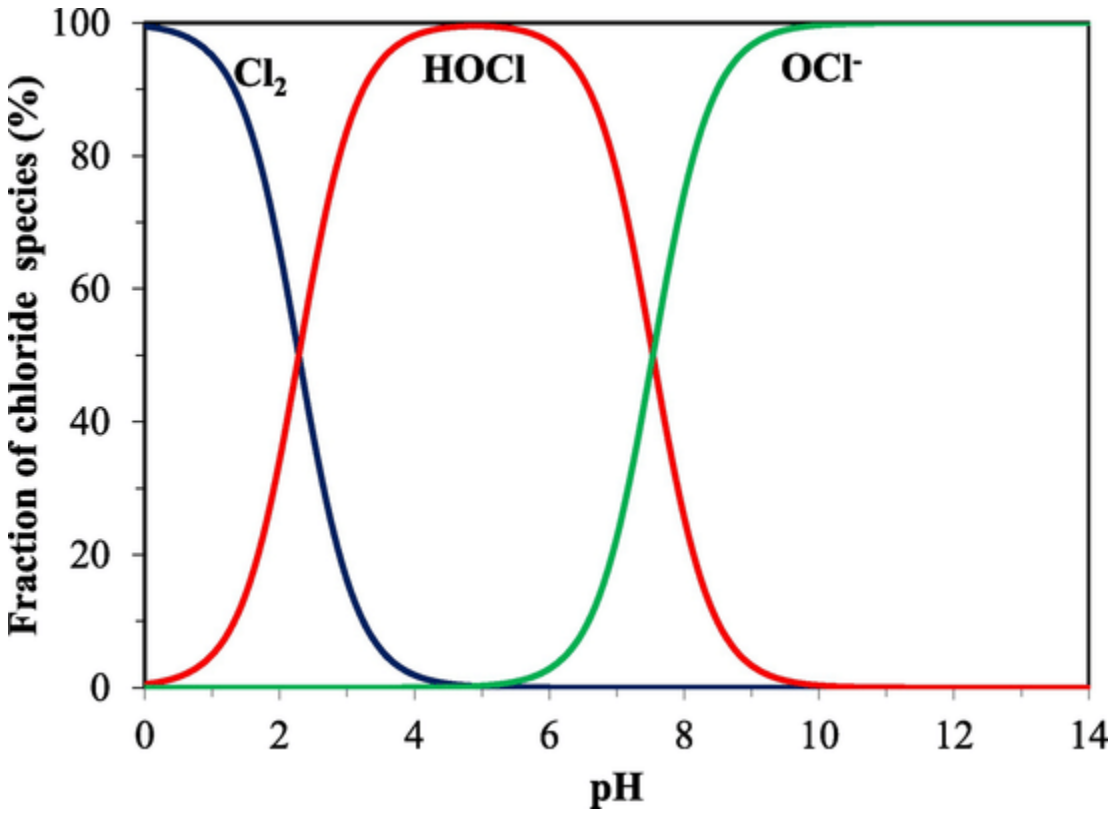
Reason (R): Chlorine dioxide is an unstable gas and does not combine with ammonia or other natural organics to produce chloramines or chloroform.

- **Correct**. Chlorine dioxide is indeed an unstable gas and does not react with ammonia or natural organics to form chloramines or chloroform, unlike free chlorine or hypochlorite.

Given the analysis, both the Assertion (A) and the Reason (R) are correct, and the Reason (R) provides the correct explanation for Assertion (A).

So, the most appropriate answer is:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)



Given below are two statements: one is labelled as Assertion (A) and the other is labelled as

Assertion (A): Marl deposits are common in lakes/ponds having high amounts of Ca2+ and excessive algal growth

Reason (R): Removal of carbon dioxide through algal action tend to cause a shift in the form of alkalinity from bicarbonate to carbonate and from carbonate to hydroxide.

In the light of the below: above statements, choose the most appropriate answer from the options given

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

ANSWER: A

Let us analyze the Assertion (A) and the Reason (R) statements:

Assertion (A): Marl deposits are common in lakes/ponds having high amounts of Ca^{2+} and excessive algal growth.

- **Correct**. Marl deposits, which are primarily composed of calcium carbonate ($CaCO_3$), often form in water bodies with high calcium concentrations and significant algal growth.

Reason (R): Removal of carbon dioxide through algal action tends to cause a shift in the form of alkalinity from bicarbonate to carbonate and from carbonate to hydroxide.

- **Correct**. When algae remove carbon dioxide during photosynthesis, it reduces the concentration of carbonic acid, causing the pH to rise. This shift

in pH changes the form of alkalinity from bicarbonate (HCO_3^-) to carbonate (CO_3^{2-}) and, at even higher pH, to hydroxide (OH^-).

Given this analysis, both Assertion (A) and Reason (R) are correct, and Reason (R) provides a valid explanation for Assertion (A).

Therefore, the most appropriate answer is:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Which of the following bacteria preferentially accumulate phosphorous from water and are used in microbial phosphorous removal process.

(2) Serrlia

(1) Achromobacter

(3) Acinetobacter

(4) Nitrobacter

ANSWER: C

The bacteria that preferentially accumulate phosphorus from water and are used in microbial phosphorus removal processes are known as **polyphosphate-accumulating organisms (PAOs)**. Among the options provided, **Acinetobacter** is known to be a PAO and is commonly used in enhanced biological phosphorus removal (EBPR) processes.

So, the correct answer is:

(3) Acinetobacter

Here's an overview of the bacteria involved in the removal of phosphorous, chloride, and other impurities:

Phosphorous Removal

Phosphorous removal is primarily achieved through the action of **Phosphate-Accumulating Organisms (PAOs)**. These bacteria, such as **Candidatus Accumulibacter phosphatis** and **Candidatus Competibacter phosphatis**, uptake and store phosphorus in the form of intracellular polyphosphates under specific anaerobic and aerobic conditions1. This process is known as **Enhanced Biological Phosphorus Removal (EBPR)**.

Chloride Removal

Chloride removal can be facilitated by **sulphate-reducing bacteria (SRB)** and **denitrifying bacteria (DB)**. These bacteria can convert chloride ions into less harmful compounds through anaerobic dechlorination processes3. Additionally, **halophilic bacteria** (salt-loving bacteria) are being explored for their potential to metabolize chloride ions under specific conditions.

Other Impurities

Various bacteria are involved in the removal of other impurities from wastewater:

- **Nitrifying bacteria** (e.g., **Nitrosomonas** and **Nitrobacter**) convert ammonia into nitrite and nitrate, which can then be further processed by denitrifying bacteria to remove nitrogen compounds.
- **Sulfur-oxidizing bacteria** (e.g., **Thiobacillus**) convert sulfides into sulfates, aiding in the removal of sulfur compounds.
- **Fermentative bacteria** break down complex organic compounds into simpler substances, aiding in the degradation of organic pollutants.

These bacteria work together in wastewater treatment plants to remove various impurities and ensure the treated water is safe for discharge or reuse.

Choose the correct statements :

(A) The BOD test is an indirect measurement of organic matter present in water

(B) Rate at which DO is consumed is directly proportional to biodegradable organic matter present at time 'r.

(C) Standard BOD curve follows zero order kinetics

(D) For one type of waste having defined BOD rate constant, ratio of ultimate BOD and BOD5 is constant

(E) Waste having same BOD, and different rate constant can have different ultimate BOD

Choose the correct answer from the options given below :

(1) (B), (C), (D) and (E) Only

(2) (A), (B), (C) and (D) Only

(3) (A), (B), (D) and (E) Only

(4) (A), (B), (C) and (E) Only

ANSWER: C

Let's analyze each statement:

(A): The BOD test is an indirect measurement of organic matter present in water.

- **Correct**. BOD (Biochemical Oxygen Demand) measures the amount of oxygen consumed by microorganisms to decompose organic matter in water. It indirectly indicates the organic matter present.

(B): Rate at which DO is consumed is directly proportional to biodegradable organic matter present at time 't'.

- **Correct**. The rate of dissolved oxygen (DO) consumption is directly related to the concentration of biodegradable organic matter.

(C): Standard BOD curve follows zero order kinetics.

- **Incorrect**. The BOD curve typically follows first-order kinetics, not zero-order. The rate of oxygen consumption decreases over time as the organic matter is degraded.

(D): For one type of waste having defined BOD rate constant, the ratio of ultimate BOD and BOD_5 is constant.

- **Correct**. For a specific type of waste with a defined rate constant, the ratio of ultimate BOD (BOD_5) to BOD at any time is constant.

(E): Waste having same BOD, and different rate constant can have different ultimate BOD.

- **Correct**. Wastes with the same initial BOD can have different ultimate BOD if they have different degradation rate constants.

Based on this analysis, the correct answer is:

(3) (A), (B), (D) and (E) Only

Which of the following is NOT a characteristic property of coagulant used during water treatment process?

(1) Non toxic nature

(2) Reduce surface charge on colloids and restrict repulsion among them

(3) Soluble in neutral pH range

(4) Trivalent cations are preferred over di and monovalent cations

ANSWER: C

Let's analyze each statement:

(1) Non-toxic nature:

- Coagulants used in water treatment must be non-toxic to ensure they do not harm human health or the environment.

(2) Reduce surface charge on colloids and restrict repulsion among them:

- This is a fundamental property of coagulants. They work by reducing the surface charge on colloids, which helps to neutralize repulsive forces and allows particles to come together and form larger flocs.

(3) Soluble in neutral pH range:

- While some coagulants are soluble in a neutral pH range, this is not a universal requirement. Coagulants can be effective across a range of pH levels, and their solubility may vary.

(4) Trivalent cations are preferred over di and monovalent cations:

- Trivalent cations (e.g., Al^{3+}, Fe^{3+}) are generally preferred over di- and monovalent cations because they are more effective in neutralizing the charge on colloidal particles and forming larger, more stable flocs.

Given this analysis, the statement that is NOT a characteristic property of coagulants used during the water treatment process is:

(3) Soluble in neutral pH range

Arrange the following phase of bioreactions as they occur from first to last in a bioreactor.

(A) Acidogenesis

(B) Methanogenesis

(C) Maturation Phase

(D) Transitional Phase

(E) Aerobic biodegradation Phase

Choose the correct answer from the options given below :

(1) (E), (A), (D), (B), (C)

(2) (E), (D), (A), (B), (C)

(3) (A), (B), (D), (E), (C)

(4) (E), (A), (B), (D), (C)

ANSWER: B

To arrange the phases of bioreactions as they occur from first to last in a bioreactor, let's consider the typical sequence of microbial processes involved in biodegradation and anaerobic digestion:

1. **Aerobic Biodegradation Phase (E)**: This phase involves the breakdown of organic matter in the presence of oxygen. It typically occurs at the beginning of the process where aerobic microbes are active.
2. **Transitional Phase (D)**: This phase represents a transition from the production of organic acids to the next step of the anaerobic digestion process, where intermediate compounds are further broken down.

3. **Acidogenesis (A)**: In this phase, anaerobic bacteria convert complex organic matter into simpler organic acids, hydrogen, and carbon dioxide. This is one of the first steps in anaerobic digestion.
4. **Methanogenesis (B)**: In this phase, methanogenic archaea convert the intermediate products, such as acetic acid and hydrogen, into methane and carbon dioxide. This is a crucial step in producing biogas.
5. **Maturation Phase (C)**: This final phase involves the stabilization and maturation of the digested sludge. It ensures that the end-products are safe and suitable for disposal or use as fertilizer.

So, the correct order is: (E), (D), (A), (B), (C)

Which of the following are common water vapour collection materials used for water vapour removal from ambient air?

(A) Calcium sulphate

(B) Silica gel

(C) Sulfuric acid

(D) Activated Alumina

(E) Anhydrous Magnesium Perchlorite

Choose the correct answer from the options given below :

(1) (A), (B) and (D) Only

(2) (A), (B) and (E) Only

(3) (A), (B), (C), (D) and (E)

(4) (A), (B), (D) and (E) Only

ANSWER: C

Common materials used for water vapor removal from ambient air include:

- **Calcium sulfate (A)**: Also known as Drierite, it is widely used as a desiccant to remove moisture from gases and air.
- **Silica gel (B)**: A popular desiccant that effectively adsorbs water vapor from the air.
- **Sulfuric acid (C)**: Used in some applications to absorb water vapor due to its high hygroscopic nature.
- **Activated Alumina (D)**: A porous material that is commonly used as a desiccant to adsorb water vapor.
- **Anhydrous Magnesium Perchlorate (E)**: Known as Dehydrite, it is used as a drying agent to remove water vapor from gases.

Given these points, the correct answer is:

(3) (A), (B), (C), (D) and (E)

JUNE 2023

Given below are two statements:

Statement I: Phosphorus cycle begins when phosphorous compounds leach from rocks and minerals over long period of time.

Statement II: Phsophorus is mainly taken up by producer organism in the organic form.

In the light of the above statements, choose the correct answer from the options given below:

1. Both Statement I and Statement II are true

2. Both Statement I and Statement II are false

3. Statement I is true but Statement II i false

4. Statement I is false but Statement II is true

ANSWER: C

Let's analyze each statement:

Statement I: Phosphorus cycle begins when phosphorous compounds leach from rocks and minerals over long period of time.

- **True**. The phosphorus cycle does indeed begin with the weathering of rocks and minerals, which releases phosphate ions into the soil and water over long periods.

Statement II: Phosphorus is mainly taken up by producer organisms in the organic form.

- **False**. Phosphorus is primarily taken up by producer organisms (such as plants and algae) in the inorganic form as phosphate ions (PO_4^{3-}), not in the organic form.

Given this analysis, the correct answer is:

(3) Statement I is true but Statement II is false

Choose the correct statement (s)

A. P occurs naturally in rocks and released in water by weathering.

B. P exist as orthophosphate and organically bound phosphate in water.

C. dissolved inorganic P is also referred as soluble reactive phosphorus.

D. P is limiting nutrient in aquatic system.

E. It is difficult to differentiate between Zooplankton-P, bacterial-P, algae-P and inorganic P.

Choose the most appropriate answer from the options given below:

1. A. B. D and E only

2. A, B, C and D only

3. A, B, C. D and E only

4. B, C and D only

ANSWER: C

Let's evaluate each statement:

A. Phosphorus occurs naturally in rocks and is released in water by weathering:

- **Correct**. Phosphorus is indeed found in rocks and minerals and is released into water bodies through the process of weathering.

B. Phosphorus exists as orthophosphate and organically bound phosphate in water:

- **Correct**. In water, phosphorus can exist in the form of orthophosphate (inorganic phosphate) as well as organically bound phosphate.

C. Dissolved inorganic phosphorus is also referred to as soluble reactive phosphorus:

- **Correct**. Dissolved inorganic phosphorus, particularly in the form of orthophosphate, is commonly referred to as soluble reactive phosphorus (SRP) because it is readily available for biological uptake.

D. Phosphorus is a limiting nutrient in aquatic systems:

- **Correct**. Phosphorus is often the limiting nutrient in freshwater systems, meaning its availability can constrain the growth of algae and other aquatic plants.

E. It is difficult to differentiate between Zooplankton-P, bacterial-P, algae-P, and inorganic P:

- **Correct**. Differentiating between the different forms of phosphorus (zooplankton-P, bacterial-P, algae-P, and inorganic P) can be challenging due to their similar chemical properties and interactions in the aquatic environment.

Given this analysis, the correct answer is:

(3) A, B, C, D and E

Standard Bio-chemical oxygen demand test is performed in 300 ml BOD bottle at

1. 25°C after 5 days
2. 15°C after 5 days
3. 20°C after 5 days
4. ambient temperature after 5 days

ANSWER: C

The standard Biochemical Oxygen Demand (BOD) test is typically performed at **20°C after 5 days** in a 300 ml BOD bottle. This temperature and duration are chosen to standardize the test conditions and ensure comparability of results.

Therefore, the correct answer is:

(3) 20°C after 5 days

March 2023

Calculate hardness in units of mg/l as CaCO3 of a ground water sample having Ca ion concentration of 100 mg/l.**

1.~5

2.~250

3.~5000

4. ~200

ANSWER: B

To calculate the hardness of the groundwater sample in mg/l as $CaCO_3$, given the Ca^{2+} ion concentration of 100 mg/l, we can use the following formula:

$$\text{Hardness (mg/l as } CaCO_3) = Ca^{2+} \text{ concentration (mg/l)} \times \frac{\text{Equivalent weight of } CaCO_3}{\text{Equivalent weight of Ca}}$$

- The equivalent weight of Ca is approximately 20 mg/mmol.
- The equivalent weight of $CaCO_3$ is approximately 50 mg/mmol.

Now, substituting the values:

$$\text{Hardness (mg/l as } CaCO_3) = 100 \times \frac{50}{20}$$

$$\text{Hardness (mg/l as } CaCO_3) = 100 \times 2.5$$

$$\text{Hardness (mg/l as } CaCO_3) = 250$$

So, the hardness of the groundwater sample in units of mg/l as $CaCO_3$ is approximately 250 mg/l.

Therefore, the correct answer is:

2. ~250

Choose the correct statements

A. Temperature has a effect in disinfection treatment of water.

B. pH controls the amount of HOCI and OCI in water.

C. Cl_2 gas used for disinfection forms HCl as a product in water.

D. Disinfection action is because of HOCI formed and not due to Cl_2 gas.

E. Combined chlorine exist in water due to combination of HOCI with NH_3 and organic compounds.

Choose the correct answer from the options given below:

1. A, C and E only

2. B, C and D only

3. A, B, C, D and E

4. A, B, C and D only

ANSWER: C

Let's evaluate each statement:

A. Temperature has an effect in disinfection treatment of water:

- **Correct**. Temperature can influence the effectiveness of disinfection processes. Higher temperatures generally enhance the disinfection rate.

B. pH controls the amount of HOCI and OCI^- in water:

- **Correct**. The pH of the water determines the ratio of hypochlorous acid (HOCl) to hypochlorite ion (OCl^-). At lower pH, HOCl is predominant, while at higher pH, OCl^- is predominant.

C. Cl_2 gas used for disinfection forms HCl as a product in water:

- **Correct**. When chlorine gas (Cl_2) is added to water, it reacts to form hypochlorous acid (HOCl) and hydrochloric acid (HCl).

D. Disinfection action is because of HOCl formed and not due to Cl_2 gas:

- **Correct**. The primary disinfecting agent in chlorination is hypochlorous acid (HOCl), which is formed when chlorine gas dissolves in water. HOCl is more effective than Cl_2 gas itself in disinfection.

E. Combined chlorine exists in water due to the combination of HOCl with NH_3 and organic compounds:

- **Correct**. Combined chlorine (chloramines) is formed when HOCl reacts with ammonia (NH_3) and organic nitrogen compounds. Chloramines are used as disinfectants, though they are less effective than free chlorine.

Given this analysis, the correct answer is:

(3) A, B, C, D and E

Choose the correct statements

A. Increase in temperature reduces DO in lake water.

B. Agricultural run-off introduces turbidity in lake water.

C. Sewage discharge increases both DO and BOD in lake water.

D. Enrichment of nutrients speed-up the aging of lakes.

E. Atmospheric exchange and photosynthesis introduces DO in lake water.

Choose the correct answer from the options given below:

1. A, C and D only
2. B, C, D and E only
3. A, B, C, D and E
4. A, B, D and E only

ANSWER: D

Let's evaluate each statement:

A. Increase in temperature reduces DO in lake water:

- **Correct**. Higher temperatures decrease the solubility of oxygen in water, thereby reducing dissolved oxygen (DO) levels.

B. Agricultural run-off introduces turbidity in lake water:

- **Correct**. Agricultural runoff often contains soil particles and other materials that increase turbidity in the water.

C. Sewage discharge increases both DO and BOD in lake water:

- **Incorrect**. Sewage discharge typically increases BOD (Biochemical Oxygen Demand) but decreases DO because the organic matter in sewage consumes oxygen during decomposition.

D. Enrichment of nutrients speed-up the aging of lakes:

- **Correct**. Nutrient enrichment, or eutrophication, accelerates the aging process of lakes by promoting excessive growth of algae and other aquatic plants.

E. Atmospheric exchange and photosynthesis introduce DO in lake water:

- **Correct**. Both atmospheric exchange and photosynthesis by aquatic plants and algae add dissolved oxygen to the water.

Given this analysis, the correct answer is:

(4) A, B, D and E only

Given below are two statements:

Statement I: Soluble phosphorus is referred to as soluble reactive phosphorus and is a limiting nutrient in aquatic systems.

Statement II: Speed of phosphorus movement between biotic and abiotic components, makes it difficult to differentiate between Zooplankton - p, bacterial-p, algae - p or even inorganic-p.

In light of the above statements, choose the most appropriate answer from the options given below:

1. Both Statement I and Statement II are correct
2. Both Statement I and Statement II are incorrect
3. Statement I is correct but Statement II is incorrect
4. Statement I is incorrect but Statement II is correct

ANSWER: A

Let's analyze each statement:

Statement I: Soluble phosphorus is referred to as soluble reactive phosphorus and is a limiting nutrient in aquatic systems.

- **Correct.** Soluble reactive phosphorus (SRP) is the form of phosphorus that is readily available for biological uptake and is often a limiting nutrient in aquatic ecosystems, controlling the growth of algae and other aquatic plants.

Statement II: Speed of phosphorus movement between biotic and abiotic components makes it difficult to differentiate between Zooplankton-P, bacterial-P, algae-P, or even inorganic-P.

- **Correct.** The rapid cycling of phosphorus between various biotic (living) and abiotic (non-living) components in aquatic systems can indeed make it

challenging to distinguish between different forms of phosphorus, such as those associated with zooplankton, bacteria, algae, and inorganic phosphorus.

Given this analysis, the most appropriate answer is:

1. Both Statement I and Statement II are correct

SEPTEMBER 2022

Choose the correct sequence of filtration processes used for sequential removal of particles from coarser (1 µm) to finer grain sizes.

A. Microfiltration

B. Ultrafiltration

C. Nanofiltration

D. Reverse Osmosis

Choose the correct answer from the options given below

1. D, C, B, A
2. A, B, C, D
3. D, C, A, B
4. A, B, D, C

ANSWER: B

Filtration Process	Particle Size Range (approximate)
Microfiltration	0.1 to 1 micrometer
Ultrafiltration	0.01 to 0.1 micrometer
Nanofiltration	0.001 to 0.01 micrometer
Reverse Osmosis	0.0001 micrometer

Given below are two statements, one is labelled as Assertion A and the other is labelled as

Assertion A: Active ingredient in most pesticides are organic chlorine compounds.

Reason R: Organo chlorines have low solubility in water, high solubility in hydrocarbon like environment and have relatively high toxicity to insects and low toxicity to humans.

In light of the above statements, choose the correct answer from the options given below

1. Both A and R are true and R is the correct explanation of A
2. Both A and R are true but R is NOT the correct explanation of A
3. A is true but R is false
4. A is false but R is true

ANSWER: A

Let's analyze both statements:

Assertion A: Active ingredients in most pesticides are organic chlorine compounds.

- **False.** While some pesticides, particularly older ones, contain organochlorine compounds (such as DDT and chlordane), many modern pesticides use different active ingredients that are not organochlorines.

Reason R: Organochlorines have low solubility in water, high solubility in hydrocarbon-like environments, and have relatively high toxicity to insects and low toxicity to humans.

- **True.** Organochlorines are characterized by their low water solubility, high solubility in organic solvents, and they are known for their effectiveness

against insects. However, their toxicity to humans and wildlife, along with environmental persistence, has led to their restriction and phase-out in many countries.

Given this analysis, the correct answer is:

BOD test is commonly used to

A. determine amount of oxygen required for biological treatment of organic matter present in waste water.

B. determine size of waste water treatment facility.

C. assess efficiency of treatment process.

D. determine compliance with wastewater discharge limits.

Choose the correct answer from the options given below:

1. D only

2. A, B and C only

3. B, C and D only

4. A, B, C and D

ANSWER: D

The Biochemical Oxygen Demand (BOD) test is a widely used method for assessing the level of organic pollution in water and has several important applications in wastewater management. Let's evaluate each statement:

A. Determine amount of oxygen required for biological treatment of organic matter present in wastewater:

- **Correct**. BOD measures the amount of oxygen required by microorganisms to break down organic matter in the water, which is essential for designing biological treatment processes.

B. Determine size of wastewater treatment facility:

- **Correct**. BOD data is used to estimate the organic load in wastewater, helping in the design and sizing of treatment facilities to handle the required capacity.

C. Assess efficiency of treatment process:

- **Correct**. BOD tests are performed before and after treatment to evaluate the effectiveness of the treatment process in reducing organic pollutants.

D. Determine compliance with wastewater discharge limits:

- **Correct**. Regulatory agencies often set discharge limits for BOD levels, and testing ensures compliance with these standards to protect water quality.

Given this analysis, the correct answer is:

4. A, B, C and D

What will be BOD5 for a wastewater sample when 21 ml of sample is diluted to 300 ml in BOD bottle to determine dissolved oxygen.

[Given: Initial DO_1 = 10 mg/l ; Five days later DO5 = 5 mg/l]

1.~5 mg/l

2.~ 25 mg/l

3.~ 71.4 mg/l

4. ~ 75 mg/l

ANSWER: C

To calculate the BOD_5 (Biochemical Oxygen Demand over 5 days) for the wastewater sample, we use the following formula:

$$BOD_5 = \frac{(DO_1 - DO_5) \times \text{Dilution Factor}}{\text{Volume of Sample}}$$

Given:

- Initial dissolved oxygen (DO_1) = 10 mg/L
- Dissolved oxygen after 5 days (DO_5) = 5 mg/L
- Volume of the sample = 21 mL
- Total volume of the dilution = 300 mL

First, calculate the dilution factor:

$$\text{Dilution Factor} = \frac{\text{Total volume}}{\text{Volume of sample}} = \frac{300}{21}$$

$$\text{Dilution Factor} \approx 14.2857$$

Now, calculate BOD_5:

$$BOD_5 = (DO_1 - DO_5) \times \text{Dilution Factor}$$

$$BOD_5 = (10 - 5) \times 14.2857$$

$$BOD_5 = 5 \times 14.2857$$

$$BOD_5 \approx 71.4 \text{ mg/L}$$

Therefore, the correct answer is:

3. ~71.4 mg/L